Build Solution

Given a use case, describe what should be considered when building a scalable solution.

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After studying this topic, you should be able to:

- Understand what scalability is and its impact on certain organizational aspects
- Identify the different factors that affect scalability and how to mitigate their impact
- Define the considerations for efficiently managing data in an environment with a large data volume
- Determine how to create a data management plan to build a scalable system
- Identify tests and processes that can be performed to evaluate and track org performance

Introduction

A system is referred to as being **scalable** when it doesn't need to make adjustments in order to operate normally when there is a **substantial increase in workload** as a result of user spikes or growth in data volume. In this topic, the considerations and recommendations for attaining scalability especially in environments with **large data volume** will be covered. Identifying the **factors** that impact scalability, defining a **data management plan**, and tracking **org performance** on a regular basis are key considerations to building a scalable solution.

Scalability Factors

Large Data Volume
When a single object contains an excessively high number of records

User Registration Rate
When a large number of users need to be created on-demand

Transaction Complexity
When a combination of automations, UI transactions, jobs, and events run together

Org Complexity
When there is a large number of users, roles, sharing rules, components, and other elements

Ownership Skew
When a single user owns a large number of records from a single object

Role Hierarchy Complexity
When there is a large number of roles assigned to users and extensive role depth

Sharing Rules
When there is a large number of sharing rules that can cause excessive sharing calculations

Org Triggers
When there is a large number of active Apex triggers in the org

Data Skew
When a single parent is associated with a large number of child records of a single object

What is Scalability

Scalability is the ability of a system to **perform efficiently** regardless of increased workload or demand of its resources. Scalability has a crucial impact on certain **organizational aspects**.

TRUST

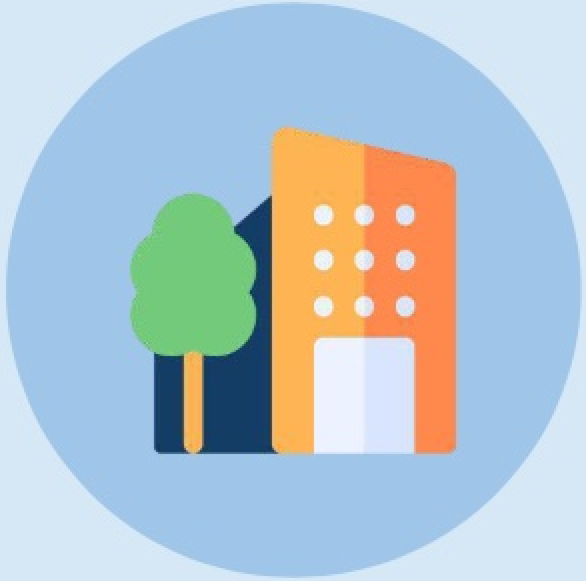
Proven **performance** and **stability** build users' trust and confidence in the integrity and services of the org.

DATA

Scalability facilitates proper data management that enables the org to **access**, **query**, or **serve** data to users in an efficient manner.

LIFESPAN

A system designed for scalability results in **better user experience**, **lower costs**, and **higher agility**, which determines the org's lifespan.



Scalability Factors and Considerations

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What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

LARGE DATA VOLUME (LDV)

When a **single object** contains an excessively **high number of records** as a result of data skewing, overly complex data structures, access rules, and other factors.

IMPACT

- Adversely affects queries, search, list views, and loading of reports
- Can cause system timeouts

MITIGATION

- Use strategic data modeling and data governance
- Use custom indexes, skinny tables, or data virtualization
- Avoid data skew
- Disable triggers and rules when loading data

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

USER REGISTRATION RATE

When there is a large number of **new users** that need to be **created** immediately in the org

IMPACT

- Can cause system timeouts

MITIGATION

- Always create the user accounts in advance if possible

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

TRANSACTION COMPLEXITY

When a **combination** of triggers, flows, UI transactions, data sync jobs, or asynchronous events **running simultaneously** in the org

IMPACT

- Can cause record locking scenarios due to concurrent transactions

MITIGATION

- Focus on low-code solutions to meet UX requirements
- Deliver pro-code solutions that leverage on existing automation capabilities and considered limitations
- Run critical batch and data sync jobs at off-peak hours

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

ORG COMPLEXITY

When there is a significantly **high number** of users in various roles, configured sharing rules, developed custom components, and installed multiple packages

IMPACT

- Recursive triggers may result in record locking
- Sharing calculations may result in slower transactions

MITIGATION

- Avoid recursive triggers

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

ROLE HIERARCHY COMPLEXITY

When there is a **large number** of users, roles, and extensive role depth

IMPACT

- Can cause data management to become more complex
- Extensive role depth will affect certain operations

MITIGATION

- Do not mirror a company's organizational role hierarchy to the org's role hierarchy
- Combine roles that need the same level of data access
- Configure the org's role hierarchy to be no more than 10 levels deep

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

SHARING RULES COMPLEXITY

When there are **redundant** sharing rules and sharing rule calculations become **excessive**

IMPACT

- Can slow down transactions when roles are changed and can impact business operations

MITIGATION

- Limit the number of ownership based sharing rules to 100 per object
- Limit the number of criteria-based sharing rules to 50 per object
- Defer sharing rule calculations when loading data

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

ORG TRIGGERS

When there is an **excessive number** of active Apex triggers for **several objects** in the org

IMPACT

- The higher the number of triggers in a large org, the more difficult it becomes to maintain and manage them

MITIGATION

- Use a trigger only if required
- Use one only trigger per object
- Bulkify helper class methods
- Use collections for *DML* statements and in *SOQL WHERE* clauses
  - Disable related triggers during data loads if possible

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

DATA SKEW

When a **single parent record** has more than **10,000 child records** within the org

IMPACT

- Slows queries, searches, list views, reports, dashboards, and sandbox refreshes

MITIGATION

- Keep child records per parent below 10,000
- Assign excess children to another parent

What Impacts Scalability

The following describes **factors** that impact scalability and the design or processes to consider in order to **mitigate their impact**.

OWNERSHIP SKEW

When a **single user** owns more than **10,000 records** of a single object

IMPACT

- Extensive sharing recalculations will occur whenever a user changes role or group membership

MITIGATION

- Distribute record ownership across a greater number of users

Large Data Volumes

Large Data Volumes

Orgs that contain **thousands of users**, **millions of records**, or consume **hundreds of gigabytes of storage** face scalability issues if not managed effectively. Considerations are stated below:

DATA MODEL DESIGN

Although scalability is about handling **workload**, how **data model is structured** in a large enterprise is critical to org performance and stability.

QUERIES AND SEARCHES

Performing queries in an org with large data volume **can consume more time**. Options are available to help **improve** the search experience.

LOADING DATA

Importing a large number of records into Salesforce demands resources. Related processes **can be deferred** to minimize consumption of resources.

DELETION AND EXTRACTION

There are recommended tools and features that help **mitigate the impact** when deleting or extracting records in a large org.

Data Model Design

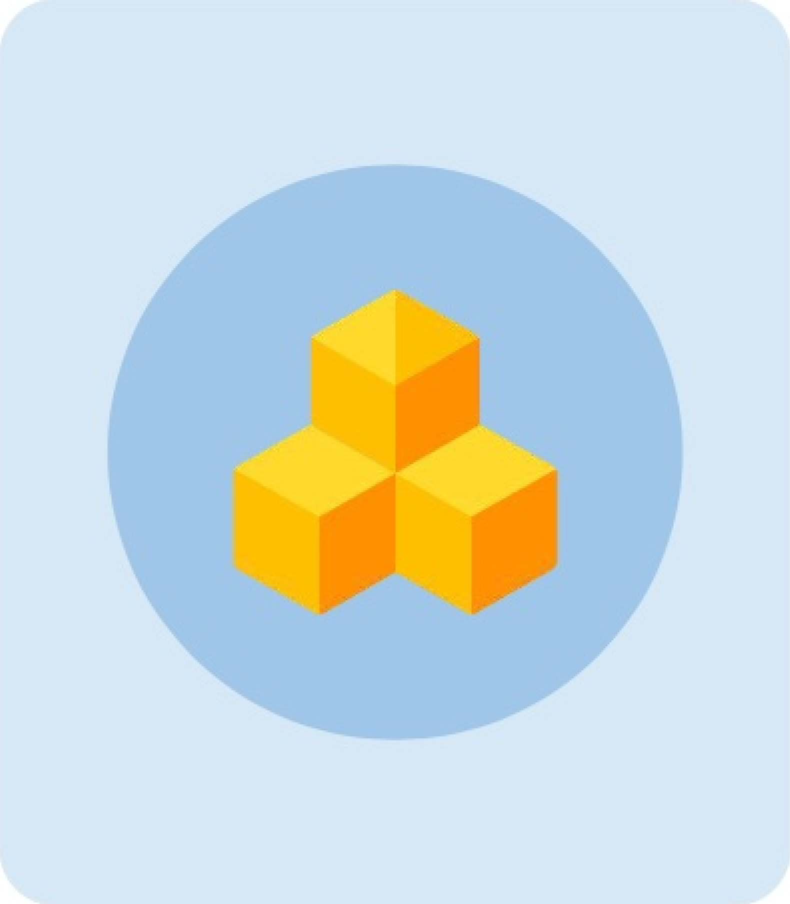
The following describes **considerations** and **recommendations** related to designing the data model especially in an environment with a large data volume.

DATA SKEW AVOIDANCE

When there are **more than 10,000 child records** that belong to the **same parent record**, it is called data skew and typically results in costly **sharing recalculations** and failed insert/update activities due to **record locking**.

DATA SKEW TYPES

- **Account data skew** - when an account has an excessive number of contacts. To avoid this, child records can be distributed across different parents systematically as they are being created
- **Ownership skew** - when a user owns a large number of records from a single object. If this cannot be avoided, sharing recalculations can be minimized by removing the user from the role hierarchy.



Data Model Design

DATA SKEW TYPES (cont'd...)

- **Lookup skew** - when there is a large number of records that are associated with a lookup object. Saving a child record also locks records that are related as lookup parents of the object. The data model should be carefully evaluated.

EXTERNAL OBJECTS USAGE

One option to consider to **reduce data volume** in the org is by using External Objects. This means storing data in an external system through **Salesforce Connect** which enables users to work with the external data from within the org. In addition, **Files Connect** can be used to access content from third-party systems.

Data Queries and Searches

The following describes **considerations** and **recommendations** related to running queries and searches especially in an environment with a large data volume.

QUERY BUILDING

Search functionality should use **selective queries** for best performance. **SOQL** can include **child-to-parent relationships** and **vice versa** in its queries. **SOSL** is optimized for performing **text searches** across **multiple** objects and fields. While **SOQL** is limited to return **50,000** records, **SOSL** is limited to **2,000** records.

QUERY OPTIMIZER

Salesforce uses a query optimizer to determine which **execution plan** performs best when handed a query based on statistics from record indexes. This feature is utilized when loading **reports** and **list views**, or when performing **SOSL/SOQL queries**.

Data Queries and Searches

BATCH APEX

Batch Apex can be used to query and process large data sets **asynchronously**. Up to **50 million records** can be handled at a time.

BULK QUERIES

The Bulk API can be used to query large data sets. Up to a total of **15GB** of data can be queried, which are split into batches of **fifteen 1-GB files**.

SKINNY TABLES

Skinny tables can be created by contacting Salesforce support to **enable faster data searching** against a table that contains a **subset of fields** of an object.

Loading Data

The following describes some processes or configurations that can be **deferred**, or performed at a later time, when **loading data** to minimize impact on org performance during the operation.

ORG-WIDE DEFAULTS

While data import of records of an object with a **Private** sharing model triggers sharing calculation, objects that use **Public Read/Write** don't.

RELATIONSHIPS

A record is locked whenever it is selected as a parent to a record when the child is being inserted or updated. Establishing the **relationships between objects/records** can be performed only after the data loading has been completed.

Loading Data

SHARING RULES

Sharing rules can be implemented after the data import has completed as each loaded record will trigger a **sharing calculation**.

AUTOMATIONS

Apex triggers (provided that a “deactivation” mechanism is implemented if in production), flows, assignment rules, and other automations **can be disabled** to minimize resource consumption during the data load operation.

Deleting and Extracting Data

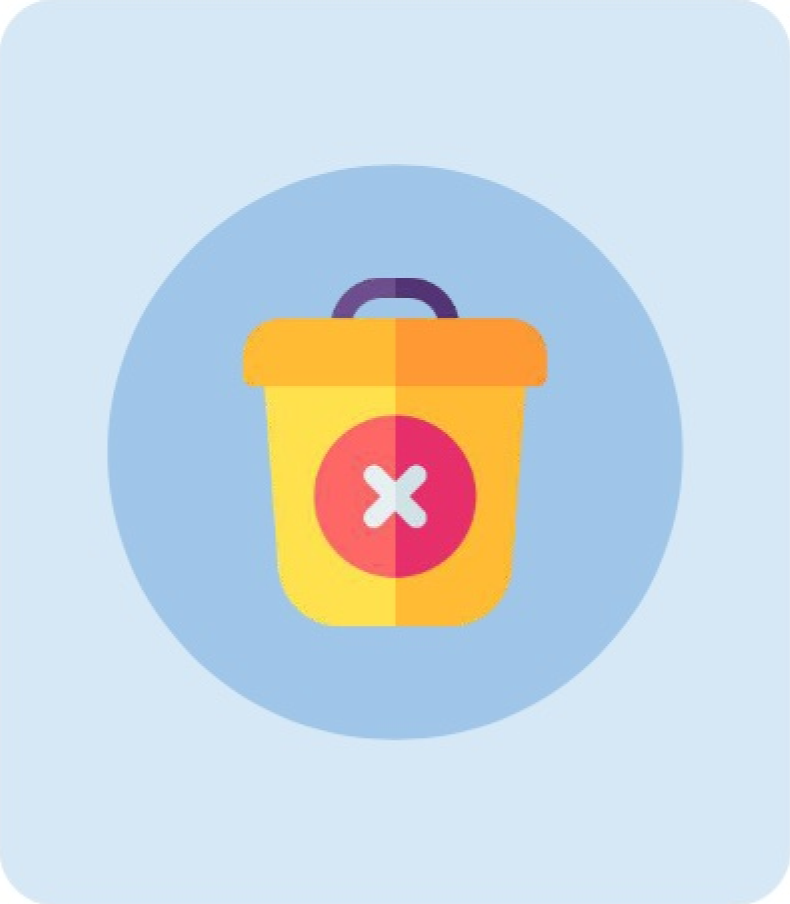
The following provides **considerations** and **recommendations** when deleting or extracting data especially in an environment with a large data volume.

BULK API USAGE

Bulk API can be used to delete more than **1 million** records. Its **hard delete** function is recommended for deleted records to bypass the **Recycle Bin**, immediately freeing up storage and excluding the records from any searches or queries.

TRUNCATION

Records of a **custom object** can be **permanently deleted** simultaneously by truncating it. This option is available in the **Object Management Settings** page of a custom object.



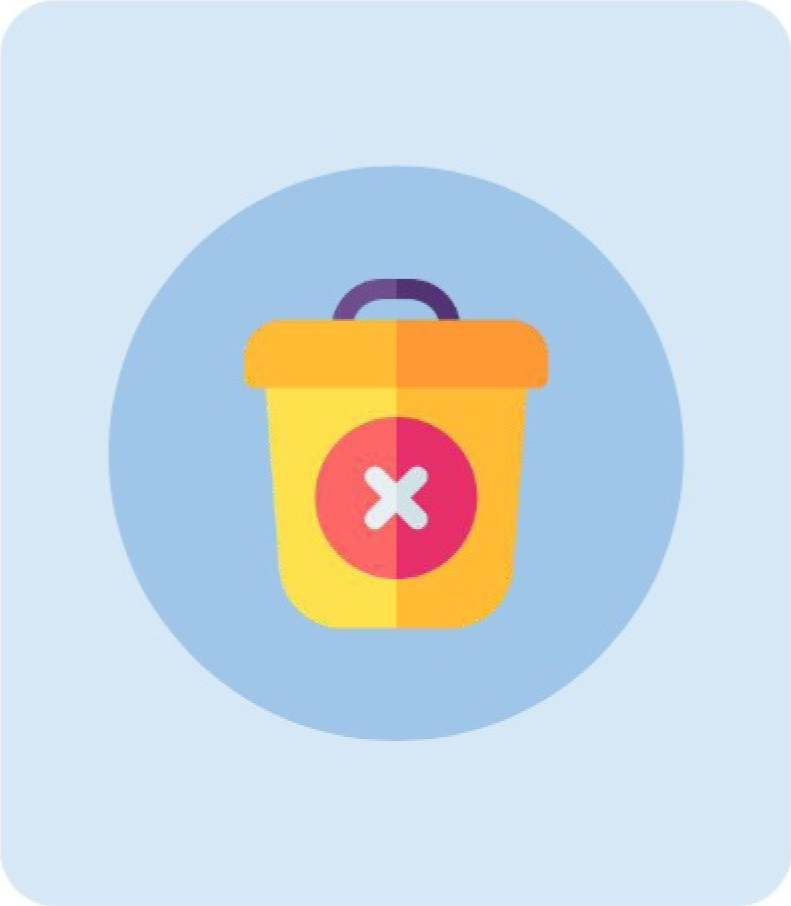
Deleting and Extracting Data

CHUNKING DATA

When extracting data using **Bulk API**, records are automatically split into chunks based on the **chunk size** (default = 100,000 records, maximum = 250,000 records).

PK CHUNKING

When breaking **extremely large data sets**, such as a table that contains more than 10 million records, into **smaller chunks** by filtering records based on **field values**, this approach may result in performance degradation or failure. In this case, PK (Primary Key) chunking can be used to automatically split bulk queries into manageable chunks based on the **record IDs** of the queried records.



Data Management

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Data Management

How data is managed can impact data integrity and user experience in the org. The following describes some **policies and processes** to ensure data is managed effectively.

DATA GOVERNANCE

Rules and **policies** should be established to ensure that the org contains customer data that is reliable and relevant to the business.

DATA STEWARDSHIP

Certain roles should be created to take on the **responsibility** of ensuring that the data governance plan is followed.

DATA MANAGEMENT STRATEGY

A data management strategy should be created to describe aspects such as how data is **handled** and what data will be **stored** in the org.

Data Governance

To ensure the **usability**, **quality**, and **compliance** of data in the org, rules and policies should be created to define **processes** and **protocols** that govern the following data-related aspects.

DATA USAGE

Rules should be established on how to **view**, **create**, and **update** data across several different sources.

DATA STORAGE

Protocols should be followed when storing data across different **cloud-based** platforms or **on-premises** storage facilities.

DATA ACCESS

Policies should govern how **external applications** and **monitoring tools** access data in Salesforce.

Data Governance

DATA SECURITY

Data should be **secure** and easily portable between systems (**data portability**). Additionally, it should be deletable without undue delay upon customer request (**right to be forgotten**)

DATA RETENTION

Data should be archived or deleted from the org in accordance with relevant **compliance** and **regulatory requirements**.

Data Stewardship

To support and ensure that the rules and policies created in accordance with the company's **data governance plan** are adhered to, the following roles should be appointed.

DATA STEWARD

A data steward is appointed to ensure **data quality** and is involved with the **processes**, **policies**, and **guidelines** related to data administration. A user in this role acts as a **liaison** between the technical (IT) side and business side of the company.

DATA CUSTODIAN

A data custodian is appointed to design, establish, or determine how data is **structured**, **stored**, **moved**, or **accessed**.

Data Management Strategy

To effectively manage data, a company should define its own data management strategy. A data management strategy should be able to **satisfy the following questions** at a minimum.

DATA TYPES

What data is **important** and **relevant** to the business and needs to be **captured** and **stored**?

STORAGE LOCATION

Where will data be stored? Will data also be stored externally using Salesforce Connect, the Heroku platform, or another storage platform?

STORAGE SIZE

What is the required storage size in order to **accommodate the data**? How fast is the data **projected to grow** in a given period of time?

Data Management Strategy

DATA MOVEMENT

Is there a requirement to **load** or **export** data to external sources? Does data move **between systems**?

DATA DUPLICATES

Is there a protocol or process in place to **manage** and **resolve** duplicate records?

DATA OWNERSHIP

Is there a **policy** that defines which **users** or **roles** should own specific **types** of business records?

Data Management Strategy

DATA SECURITY

Is there a security model in place that defines **who** can access **what** data and the **operations** they are authorized to perform on that data?

DATA ENCRYPTION

Does data need to be encrypted when stored (**at rest**) in the org or when included in a transaction (**in transit**)?

DATA DELETION

Is there a **protocol** for when and how to **archive** or **delete** data that is redundant, no longer necessary, or has been requested for deletion?

Testing Platform Performance

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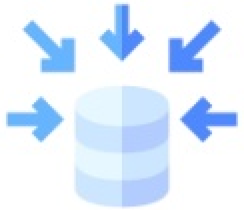
Testing Platform Performance

There are procedures that can be performed in order to **determine the behavior** and **test the performance** of an org.

SCALABILITY TESTING

This test is used to evaluate org behavior when **changes** in **hardware**, **database**, and **network** are made to meet increased demand in terms of **user traffic**, **data volume**, and **transactions**.



LOAD TESTING

This type of test evaluates the **response time** of the org when there is an **increased number of users** accessing and consuming resources of the org.

DATA MONITORING

This process tracks org performance by monitoring the **activities** that users perform or event trends to determine areas of **high resource consumption** and **abnormal behavior**.

Scalability Testing

Scalability testing can be used to determine the maximum capacity of an org. These **performance attributes are monitored** when application hardware, database, and network are changed.

RESPONSE TIME

For example, the **time duration** required to receive a response after a request is submitted

THROUGHPUT

For example, the **number of requests** that can be processed successfully in a given period of time

CPU USAGE

For example, the **amount of time** that the server CPU will require to complete a specific task

Scalability Testing

MEMORY USAGE

For example, the **amount of memory** that will be consumed when performing a specific task

NETWORK USAGE

For example, the **amount of data** that can be transported across the network in response to requests

Load Testing

Load testing focuses on **response times** when there is a **realistic increase** in the number of users. The **following attributes** are monitored when performing this test.

PEAK PERFORMANCE

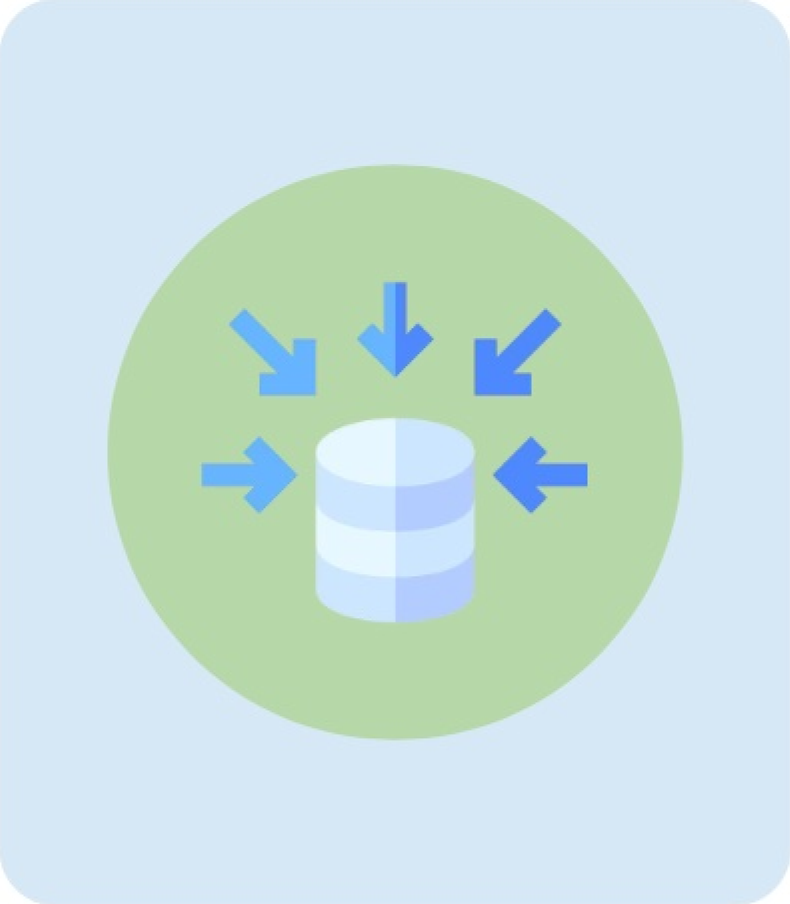
Performance is tested by **simulating load** that is experienced during peak hours or peak concurrent users.

SERVER THROUGHPUT

The **number of requests** that were processed by the server per unit of time is monitored.

RESPONSE TIME

The **round-trip time** is tested under different load levels but below the breaking point of the org.



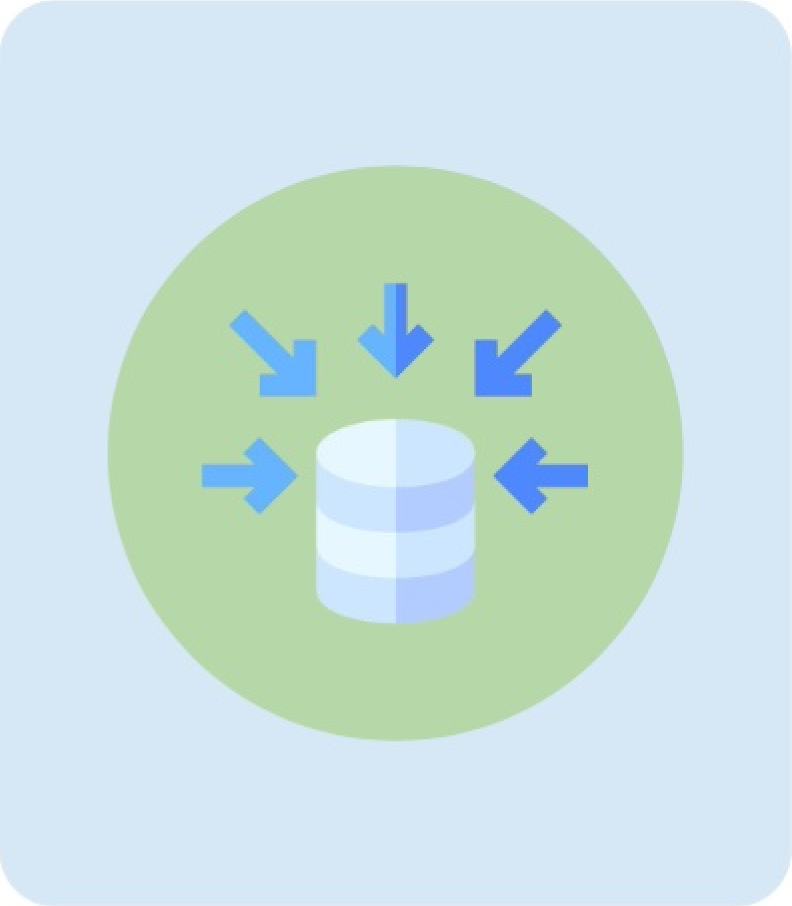
Load Testing

HARDWARE ROBUSTNESS

Hardware such as org **CPU**, **memory**, and **database** are monitored under various loads to determine its robustness.

NUMBER OF APPLICATIONS

Applications are being run to determine the **serving capacity** of the org without degrading performance.



Data Monitoring

Monitoring **events** and **user activities** helps to understand how data is being accessed or used in the org and identifies **peak data usage** or **abnormal behavior**.

EVENTS

The following lists some **events** that should be **monitored**.

- User logins and logouts
- Record searches, report exports
- Web clicks, page views, performance, errors
- Visualforce page loads, API calls, Apex executions

TOOLS

Salesforce provides **monitoring tools** in Setup such as the System Overview page, Storage Usage, Lightning Usage App, and more. **Real-Time Event Monitoring** can also be used to monitor events such as which users viewed which data and when, where data was accessed from, etc.

Use Cases

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Scenario 1

SCENARIO

A large company in the logistics industry works with millions of records on a daily basis. The company uses Salesforce to store and manage customer data, products, orders, packaging, deliveries, logs, and several other types of records that are accessed and processed by users from different departments in their organization across different regions. As the business grows, its Salesforce org experiences rapid growth in data, and users are starting to experience slow performance and encounter request timeouts that adversely impact their productivity.

Solution 1

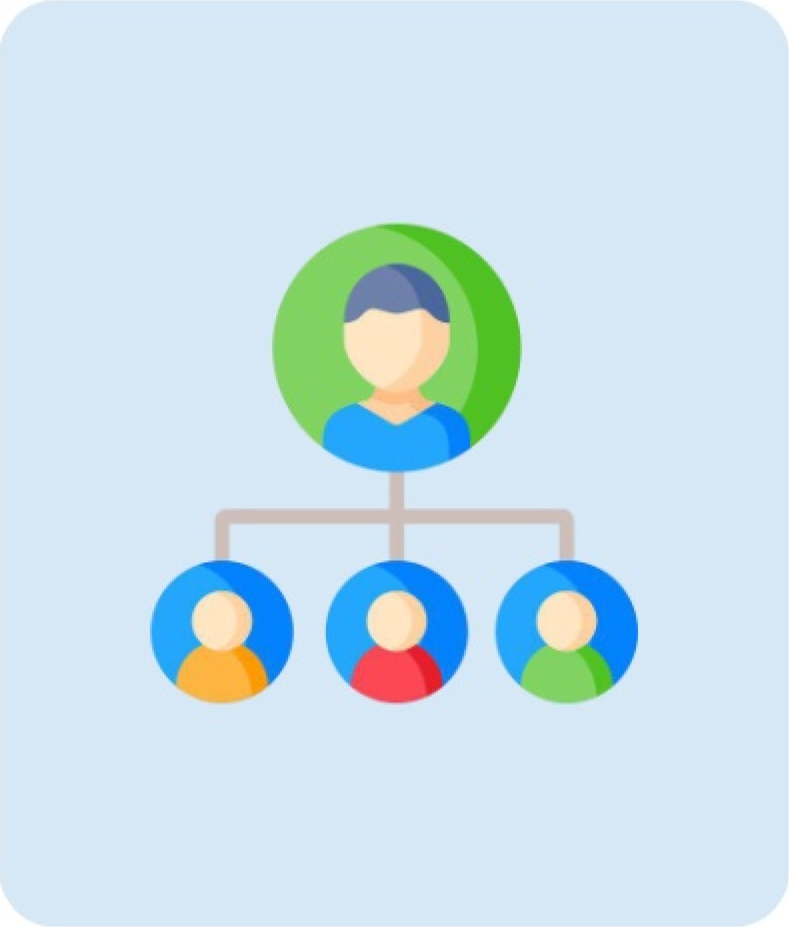
SOLUTION

To enhance performance within an organization managing a large data volume, several optimization considerations are critical for improving performance and ensuring system scalability. One such consideration involves avoiding data skew, which occurs when more than 10,000 records are related to a single parent. Operating under these conditions often leads to record locking and failed operations. Another consideration is the optimization of queries through the utilization of selective queries, Batch Apex, and Bulk API to process large amounts of data. Additionally, creating a skinny table can enhance the search experience.

Scenario 2

SCENARIO

Within a Salesforce org, the role hierarchy has been defined to mirror the organizational structure of a corporate enterprise, which is composed of several levels of departments and sub-departments. Sharing rules have also been created to define which users from the various departments have access to specific types of records. A department was restructured to focus on a specific business service, necessitating an update to the role hierarchy. However, as roles were created and updated, the organization encountered slow performance and extended response times.



Solution 2

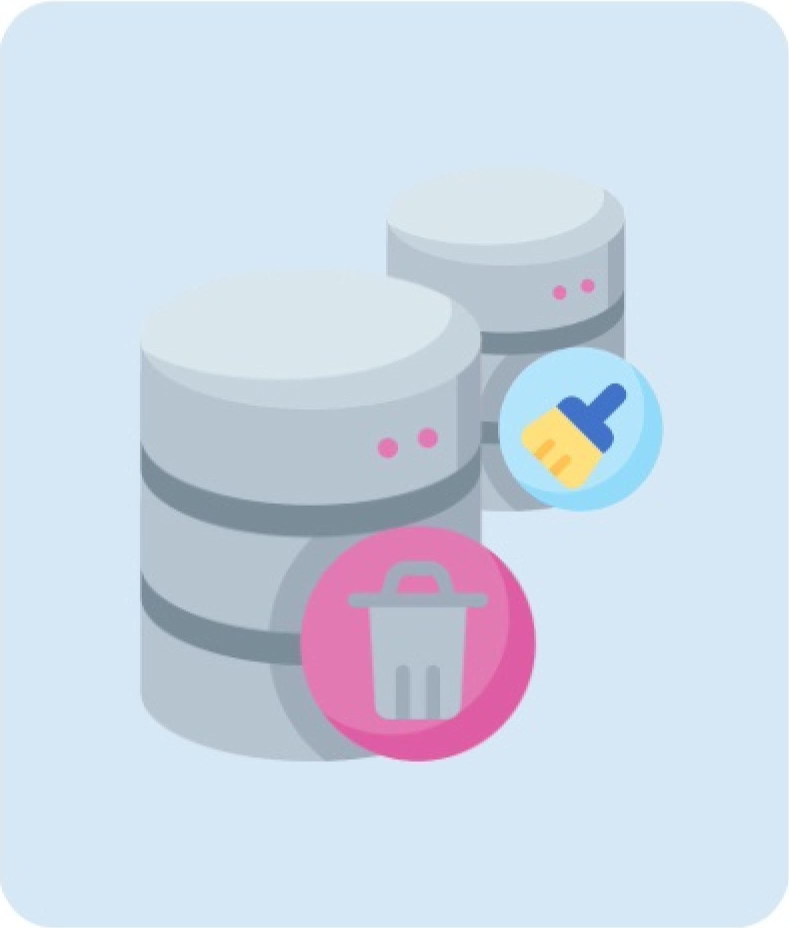
SOLUTION

The role hierarchy within an org should not mirror that of the business, but rather consist of roles created and structured according to record access requirements or the company's data management strategy. This approach helps avoid redundant roles and reduces the total number of existing roles. Salesforce recommends that role hierarchies be kept under 10 levels, as increasing role depth affects the performance of certain operations within the organization. For example, greater depth results in more costly sharing calculations when a user role is modified.

Scenario 3

SCENARIO

A company is implementing a massive data cleanup and has backed up hundreds of thousands of records in an on-premises storage facility. The subsequent step involves the Salesforce administrator deleting the records from the organization. Going forward, the company also seeks to improve workload handling by reducing the volume of data stored within its Salesforce organization.



Solution 3

SOLUTION

In this case, the records that need to be deleted can be hard-deleted using the Bulk API. Hard-deleted records bypass the Recycle Bin and are permanently and immediately deleted from the organization. Soft-deleted records are merely flagged as deleted and remain accessible via the Recycle Bin for up to 15 days. Consequently, they continue to consume storage space and have to be excluded from queries. To minimize data storage, one option is to store a significant volume of data on an external platform and access it within the organization using Salesforce Connect. With this solution, external data does not consume organizational storage, and users can access or process data in real-time.

Learn More

- [Understand Scalability at Salesforce]
- [Manage Data at Scale with Salesforce]
- [Test Platform Performance at Scale]
- [Large Data Volumes]
- [Monitor Your Organization]